ECLAIR: Anytime-Valid Certification of LLM-Generated Constraint Models with Budget-Aware Probe Routing

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Abstract

Large language models translate natural-language problem descriptions into formal constraint models, but they lack the mathematical guarantees required for formal correctness, and no ground truth exists against which a generated model can be checked — the oracle problem. Existing verification approaches are heuristic detectors: they report empirical detection rates but provide no statistical guarantee on the model one ultimately accepts, no principled stopping rule, and no way to decide which test to run next under a compute budget. We introduce ECLAIR, a sequential, anytime-valid, budget-aware certification layer. Each candidate model carries an e-process — a nonnegative supermartingale wealth process — betting against the hypothesis that the candidate is faithful, fed by three tiers of imperfect probes: exhaustively enumerated micro-instances, provably valid metamorphic relations, and cross-model majority checks. Conservative calibration of the probes' null alarm rates preserves validity under the composite null, and a Kelly-style rule routes solver time to the candidate–probe pair with the highest expected evidence growth per second. Ville's inequality bounds the probability of ever falsely rejecting a faithful model by $\boldsymbol{\alpha}$ under optional stopping, and e-BH selection controls the false discovery rate over the whole generate–test–repair tournament. On a mutation testbed of five problem families with real CP-SAT probes, the empirical false-rejection rate is zero at every nominal level while **87–94%** of injected faults are detected; an end-to-end experiment with two LLM families reproduces the guarantee on real generated models.

Keywords: constraint modelling, large language models, e-processes, anytime-valid inference, solver-in-the-loop verification, false discovery rate

1 Introduction

Constraint programming's long-standing adoption bottleneck is the expertise barrier: translating a problem description held in natural language into a formal model. Large language models (LLMs) now cross that barrier routinely — generating CPMpy, MiniZinc, or MILP models from plain-English specifications [1–3] — but they do so without any intrinsic guarantee of correctness. A generated model that compiles and solves is not necessarily the model that was asked for: a misread coefficient, a dropped constraint family, or a flipped inequality produces a plausible artifact that returns confidently wrong optima. Because the specification exists only in natural language, there is no oracle to compare against; this is the central verification difficulty the field currently faces.

The question a practitioner asks of any verification layer is: *what guarantee do I get on the model I finally accept?* Existing answers are heuristic. Detection frameworks based on metamorphic relations or mutation-style testing report empirical detection rates, but offer no control over the probability of certifying an unfaithful model, no principled rule for when enough testing has been done, and no guidance on which of many possible tests — differing wildly in cost and power — to run next under a finite solver budget. Fixed-sample statistical tools do not repair this: a generate–test–repair loop inherently peeks at accumulating evidence and adapts, which invalidates fixed-sample p-values and split-conformal calibration alike.

This paper introduces **ECLAIR** (E-process Certification of LLM-generated constraint models with Adaptive Instance Routing), a certification layer built on the game-theoretic statistics of e-processes [4–6]. The design follows three observations.

First, *optional stopping is the workflow.* E-processes are designed for continuous monitoring: rejecting whenever the wealth process crosses $1/\alpha$ controls the type-I error at α uniformly over all stopping times (Ville's inequality [7]), so the loop may stop the moment evidence suffices — a direct compute saving — and may resume testing at any time without penalty.

Second, *probes are heterogeneous, imperfect, and dependent.* A brute-forced micro-instance is a near-perfect oracle but expensive; a metamorphic relation is provably false-alarm-free but weak; a cross-model agreement check is cheap, noisy, and entangled with the candidate pool. E-values multiply across adaptively chosen, dependent probes as long as each factor is a valid e-variable conditionally on the past, and we show that a conservative upper confidence bound on each probe family's null alarm rate — rather than an unattainable exact null distribution — suffices for validity under the composite null "this candidate is faithful."

Third, *the budget question is a portfolio question.* Choosing which (candidate, probe) pair to run next to maximize expected log-evidence growth per solver-second is Kelly-optimal evidence acquisition [6, 8]; because the routing decision is measurable with respect to the past, it affects power only, never validity.

Contributions.

(i) A formal certification protocol for LLM-generated constraint models: per-candidate e-processes over a three-tier probe pool, with hard certificates (probes whose false-alarm probability is exactly zero by proof) embedded as infinite e-values (Section 4).

(ii) Validity theory for the composite null: conservative calibration preserves the supermartingale property (Lemma 1), yielding anytime-valid false-rejection control under arbitrary predictable routing (Theorem 2), false discovery rate control over the full generate–test–repair tournament via e-BH [9] (Proposition 3), and a budget bound on the expected solver time to reject an unfaithful candidate (Proposition 4). (iii) A verified probe library for five problem families — knapsack with conflicts, set cover, graph colouring, generalized assignment, and on-time job selection — including per-family metamorphic relations proved valid at specification level (Proposition 5). (iv) An open implementation on CPMpy [10] and OR-Tools CP-SAT [11] and a three-part empirical study (Section 6): a mutation testbed in which validity holds exactly at every nominal level while 87–94% of injected faults are caught; ablations quantifying the contribution of each probe tier, including an empirical demonstration that pool-entangled probes consume the conservative validity margin; and an end-to-end experiment with two LLM families in which every crash-free generated model is certified without a single false rejection.

2 Related work

LLMs for constraint modelling.

In-context learning pipelines translate natural-language descriptions into CPMpy models with retrieval and self-refinement [1]; benchmark suites evaluate such systems on CSPLib-derived corpora [12]. Agentic systems iterate modelling with solver feedback: CP-Agent [3] maintains a persistent Python session around CPMpy, and MCP-Solver [13] exposes constraint solvers to LLMs through a model context protocol. ConstraintLLM [2] targets industrial-level CP with a neuro-symbolic architecture. All these systems *generate*; ECLAIR is complementary — it decides, with a guarantee, whether to *trust* what was generated, and can wrap any of them.

Verifying generated models.

Two recent lines attack verification directly: falsification frameworks deriving provably valid metamorphic relations from duality and monotonicity theory, in which a violated relation certifies unfaithfulness, and agent-based frameworks that compose test cases and apply optimization-oriented mutation testing.[1] ECLAIR subsumes the binary verdicts of such detectors as the special case of an infinite e-value, and adds what they lack: a guarantee on the accepted model, a stopping rule, FDR control across repairs, and budget-aware test selection.

Anytime-valid inference.

E-values and e-processes provide measures of evidence valid at arbitrary stopping times [4, 5, 14]; safe tests and the GROW criterion connect optimal evidence growth to log-optimal (Kelly) betting [6, 8]; betting-based confidence sequences supply the adaptive wagering machinery [15]; and the e-BH procedure controls FDR under arbitrary dependence [9]. ECLAIR ports this toolkit into constraint-model verification, where, to our knowledge, sequential anytime-valid certification has not been attempted.

[1] Full citations pending verification.

Conformal approaches to LLM output control assume exchangeability between calibration and test data and are fixed-sample; both assumptions fail in a sequential generate–test–repair loop.

3 The certification problem

A specification s is a natural-language description of a parameterized combinatorial optimization problem, together with an instance distribution $\mathcal{D}$ over parameter values. A *candidate* m is an executable function mapping an instance $I \sim \mathcal{D}$ to a formal model $m(I)$ whose exact optimum we write $v_m(I) \in \mathbb{Z} \cup \{\text{INFEAS}\}$. The (unknown) faithful value $v^*(I)$ is the optimum of the problem s actually describes. Candidate m is *faithful* if $v_m(I) = v^*(I)$ for $\mathcal{D}$-almost-every I.

Given candidates $m_1, \ldots, m_k$ produced by an LLM ensemble, a compute budget B (solver-seconds), and a risk level α, the certifier must decide which candidates to reject, which (if any) to certify, or whether to abstain — without access to v^*. The null hypothesis attached to each candidate is

$$H_0(m): \quad m \text{ is faithful.}$$

$H_0(m)$ is *composite*: faithfulness does not determine the law of any probe outcome, which also depends on the instance generator, on checker noise, and (for cross-model probes) on the composition of the surviving pool. All guarantees below therefore hold uniformly over the null class, which we make precise through conditional bounds on probe alarm rates.

4 The ECLAIR framework

4.1 Probe tiers

A *probe* is a randomized procedure that consumes solver time and returns a binary alarm $X \in \{0, 1\}$ against a designated candidate. ECLAIR uses three tiers.

Tier A: micro-instance oracles.

Draw an instance small enough for exhaustive enumeration of the specification's decision space using an executable feasibility checker, and alarm iff the candidate's exact optimum differs from the enumerated optimum. With an exact checker the null alarm probability is zero; in general the checker itself is imperfect (in a fully LLM-generated pipeline it is LLM-compiled and validated by cross-agreement), so tier A is treated as a calibrated noisy probe.

Tier B: metamorphic relations (hard certificates).

A metamorphic relation (MR) transforms an instance I into I' such that the specification provably implies a relation between $v^*(I)$ and $v^*(I')$ — e.g. enlarging a knapsack capacity cannot decrease the optimum, deleting a graph edge cannot increase the chromatic number, consistently permuting item data leaves the optimum unchanged.

The probe solves the *candidate* on both sides exactly and alarms iff the relation is violated. For a faithful candidate the alarm probability is exactly zero: both solves return true optima of instances of the true specification, and the relation holds for the specification by proof. A tier-B alarm is therefore a *hard certificate* of unfaithfulness.

Tier C: cross-model majority checks.

Solve the designated candidate and up to three surviving partners on a fresh mid-size instance (beyond enumeration reach — this tier's purpose); alarm iff at least two partners agree on a value and the candidate deviates from that majority. Tier C is cheap and noisy, and its null alarm rate depends on the pool: a faithful candidate alarms when a wrong majority forms. This entanglement is handled by calibration (Section 4.2) and is quantified empirically in Section 6.3.

4.2 Betting and conservative calibration

Let $\mathcal{F}_{t-1}$ denote everything observed before the t-th probe. For a tier-B probe the e-factor is $e_t = 1$ on pass and $e_t = \infty$ on alarm. For tiers A and C, ECLAIR bets a Bernoulli likelihood ratio

$$e_t \;=\; \left(\tfrac{p_1}{\bar{p}_0}\right)^{X_t} \left(\tfrac{1-p_1}{1-\bar{p}_0}\right)^{1-X_t}, \tag{1}$$

where $\bar{p}_0$ is a *conservative* (upper confidence) bound on the tier's null alarm rate and $p_1 \in (\bar{p}_0, 1)$ a working detection rate; both come from a held-out calibration corpus of known-faithful models and screened mutants, with $\bar{p}_0$ the Clopper–Pearson upper bound [16] at confidence 97.5%. Only $\bar{p}_0$ matters for validity; p_1 tunes power.

Lemma 1 (Conservative bets are safe) *Let $X \in \{0,1\}$ satisfy $\Pr(X = 1 \mid \mathcal{F}_{t-1}) \le \bar{p}_0$ almost surely under H_0, and let e be given by (1) with $0 < \bar{p}_0 < p_1 < 1$. Then $\mathbb{E}[\,e \mid \mathcal{F}_{t-1}\,] \le 1$ under H_0.*

Proof With $q = \Pr(X{=}1 \mid \mathcal{F}_{t-1})$, $\mathbb{E}[e \mid \mathcal{F}_{t-1}] = q\,\frac{p_1}{\bar{p}_0} + (1-q)\,\frac{1-p_1}{1-\bar{p}_0}$, which is affine and increasing in q because $p_1/\bar{p}_0 > 1 > (1-p_1)/(1-\bar{p}_0)$. At $q = \bar{p}_0$ it equals $p_1 + (1-p_1) = 1$; hence for $q \le \bar{p}_0$ it is at most 1. □

The candidate's wealth process is the running product $E_t(m) = \prod_{s \le t,\, s \to m} e_s$ over the probes routed to m, with $E_0(m) = 1$.

4.3 Budget-aware routing

At each step ECLAIR selects

$$(m,p)^\star \;=\; \arg\max_{(m,p)} \frac{\mathbb{E}\big[\log e \;\big|\; q_m\big]}{c(p)}, \tag{2}$$

Algorithm 1 ECLAIR

Require: spec s, candidates $m_1{:}k$, budget B, level α, calibrated $(\bar{p}_0, p_1, \delta_B, c)$ per tier

$E(m_i) \leftarrow 1$, $q_{m_i} \leftarrow q_{\text{prior}}$ for all i
while $B > 0$ **and** some candidate alive **do**
  $(m, p) \leftarrow$ maximizer of (2) over alive candidates and tiers
  run probe p on m; observe alarm X, measured cost c; $B \leftarrow B - c$
  if p is tier B **and** $X = 1$ **then** reject m ▷ hard certificate
  else
    $E(m) \leftarrow E(m) \cdot e(X)$ by (1); update q_m by Bayes
    if $E(m) \geq 1/\alpha$ **then** reject m
    end if
  end if
  optionally: repair rejected m via counterexample; add as fresh e-process
end while
return e-BH rejection set at level α; certified pick $\arg\min_{\text{survivors}} E(m)$, or abstain with the evidence ledger

the candidate–probe pair maximizing expected log-wealth growth per solver-second: Kelly-optimal evidence acquisition in the sense of the GROW criterion [6, 8]. Here q_m is a running (Bayes) posterior that m is unfaithful, maintained with the same calibrated rates and used *only* for routing; $c(p)$ is the tier's calibrated mean cost. For tiers A/C the numerator is $q_m G_1 + (1 - q_m) G_0$ with G_1, G_0 the expected log-factors of (1) under alarm rates $p_1, \bar{p}_0$; for tier B it is $q_m\, \delta_B \big(\log(1/\alpha) - \log E_t(m)\big)$, the calibrated hard-certificate rate times the remaining log-gap to the rejection threshold. Because (2) is $\mathcal{F}_{t-1}$-measurable, routing affects power only — validity is untouched (Theorem 2).

4.4 Decisions, repair, and abstention

Candidate m is *rejected* the first time $E_t(m) \geq 1/\alpha$ (or on a tier-B alarm). A rejected candidate may be *repaired*: the violating probe is fed back to the LLM as a counterexample and the repaired candidate enters the pool as a fresh e-process. When the budget is exhausted (or one candidate's survival dominates), the terminal e-values of *all* candidates ever considered enter the e-BH procedure [9] at level α; among Ville-survivors the candidate with the smallest e-value is the certified output, together with its evidence ledger. If no candidate survives, ECLAIR *abstains* and returns the ledger — a structured explanation of which probes killed every reading of the specification, which doubles as a signal for conversational re-elicitation. Algorithm 1 summarizes the loop.

5 Guarantees

Assumption 1 Under $H_0(m)$: tier-B probes alarm with probability zero, and each tier-A/C probe routed to m alarms with conditional probability at most its calibrated bound $\bar{p}_0$, given $\mathcal{F}_{t-1}$.

The tier-B half of Assumption 1 is a theorem for the probe library of Section 6 (Proposition 5); the tier-A/C half is what conservative calibration targets, and its empirical status — including where it becomes delicate — is examined in Section 6.3.

Theorem 2 (Anytime validity) *Under Assumption 1, for any predictable routing rule, any repair policy, and any stopping rule,*

$$\Pr_{H_0(m)}\big(\exists\, t:\ E_t(m) \geq 1/\alpha\big) \ \leq\ \alpha.$$

Proof Each factor contributed to $E(m)$ is nonnegative and, by Lemma 1 (tiers A/C) or by the zero-alarm property (tier B, factor $\equiv 1$ a.s. under H_0), satisfies $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] \leq 1$; predictability of the routing rule makes "which probe, on which candidate" a $\mathcal{F}_{t-1}$-measurable choice, so the tower property gives $\mathbb{E}[E_t(m)] \leq 1$ and $E_t(m)$ is a nonnegative supermartingale. Ville's inequality [7] concludes. □

Proposition 3 (Tournament FDR control) *Let $\{E(m)\}$ be the terminal e-values of all candidates considered across all generation and repair rounds. Applying e-BH at level α controls the false discovery rate of the rejected set at α, regardless of the dependence induced by shared probes, common instances, pool entanglement, or data-driven repairs.*

Proof Each $E(m)$ is a valid e-value for $H_0(m)$ by Theorem 2 (optional stopping preserves $\mathbb{E} \leq 1$). The e-BH procedure controls FDR for arbitrarily dependent e-values [9]. □

Proposition 4 (Budget to rejection) *Fix an unfaithful candidate m and a routing rule under which the probes applied to m satisfy $\mathbb{E}[\log e_t \mid \mathcal{F}_{t-1}] \geq \Delta > 0$ and $|\log e_t| \leq L$. Then the number τ of probes until rejection satisfies $\mathbb{E}[\tau] \leq \big(\log(1/\alpha) + L\big)/\Delta$.*

Proof $M_t = \log E_t(m) - t\Delta$ is a submartingale minorant; optional stopping for the bounded-increment stopped process at $\tau \wedge n$ gives $\Delta\, \mathbb{E}[\tau \wedge n] \leq \mathbb{E}[\log E_{\tau\wedge n}] \leq \log(1/\alpha) + L$, and monotone convergence lets $n \to \infty$. □

Proposition 4 is the quantitative content of "stop the moment evidence suffices": the budget scales as $\log(1/\alpha)$, so tightening the guarantee is cheap — consistent with the flat detection curves across α observed in Section 6.2.

Proposition 5 (Probe-family validity) *For the five problem families of Section 6, the implemented metamorphic relations — capacity/deadline/cost monotonicity, structural relaxation (conflict, edge, or coverage-row deletion), restriction (edge addition), and consistent data permutation — have null alarm probability exactly zero when both sides are solved exactly.*

Proof sketch Each relation is proved at specification level; exact solves transfer it to any faithful candidate. Monotone relaxations follow from feasible-set inclusion (e.g. enlarging a knapsack capacity or deleting a conflict pair only enlarges the feasible set of the specification, so the maximum cannot decrease); objective monotonicity follows from pointwise dominance; permutation invariance follows from the specification's symmetry under consistent relabeling of items, sets, vertices, or jobs. The one delicate case is deadline extension in on-time job selection, where the constraint *indices* shift with the deadline order: there, validity follows because the earliest-due-date feasibility conditions are equivalent to true single-machine schedulability of the selected set, and schedulability is monotone in deadlines. Full proofs accompany the probe library. □

Remark 1 Assumption 1 for tier C deserves emphasis: the null alarm event "a wrong majority forms among partners" depends on the surviving pool, so its conditional probability is bounded uniformly only if calibration reflects the pool compositions actually encountered. Section 6.3 shows empirically that this is not pedantry — the conservative margin visibly erodes exactly, and only, when tier C is made the sole evidence source.

6 Experimental evaluation

Implementation.

All experiments run on CPMpy [10] with the OR-Tools CP-SAT solver [11]. Five problem families are implemented, each with a reference model builder, an *independent* pure-Python specification checker (the tier-A enumeration oracle), micro/mid instance generators, and the family's metamorphic relations of Proposition 5: 0/1 knapsack with pairwise conflicts, minimum-cost set cover, graph colouring (minimizing the number of colours), generalized assignment, and weighted on-time job selection on a single machine. Probe costs are measured wall-clock solver-seconds. Code, calibration corpora, all raw results, and the cached LLM generations are provided as supplementary material.

Unfaithful candidates by functional mutation.

To measure error rates one needs ground truth, so the first two studies replace the LLM with *functional corruptions* of the reference builder — systematic, instance-generic errors of the kinds LLM modellers actually make: a bound misread by a constant, one vector entry corrupted positionally, a data-wiring misalignment (vector rotation), a dropped structural row or half-family, a corrupted cost-matrix entry, and a flipped comparison operator in one constraint family. Mutants are screened for semantic difference from the reference (equivalent mutants are resampled), the standard mutation-testing practice.

Calibration.

Betting parameters are calibrated on two held-out families (knapsack with conflicts, graph colouring) and evaluated on the three *disjoint* remaining families — so every result below also tests whether conservative calibration transfers across problem families. Calibration measured: tier A, zero false alarms in 240 trials ($\bar{p}_0 = 0.0153$),

Table 1 Mutation testbed (100 replications/policy, $\alpha = 0.05$, real CP-SAT probes; calibration and evaluation families disjoint). False rejection is over faithful candidates ($n \approx 300$ per policy); detection over mutants; kill cost is mean solver time spent on a rejected mutant; pick accuracy is the probability the certified output is faithful.

Routing	False rej.	Detection	Kill cost	Pick acc.	e-BH FDR
Kelly (ECLAIR)	0.0000	0.900	20 ms	0.890	0.0
round-robin	0.0000	0.874	29 ms	0.950	0.0
cost-blind	0.0000	0.943	23 ms	0.959	0.0

detection 0.59, mean cost 5.2 ms; tier B, zero false alarms in 240 trials (asserted, per Proposition 5), hard-certificate rate $\delta_B = 0.042$, cost 6.3 ms; tier C, null alarm rate 0.117 ($\bar{p}_0 = 0.164$), detection 0.57, cost 12.4 ms.

6.1 Mutation testbed: validity, power, and routing

Pools of $k = 6$ candidates (each faithful or a screened mutant with probability 1/2) are certified at $\alpha = 0.05$ under a shared budget of 640 ms of measured solver time per pool, 100 replications per routing policy with identical pools across policies. Table 1 reports the results.

Validity holds exactly — not one of ∼900 faithful-candidate exposures across the three policies was rejected — despite the bets being calibrated on different problem families than those evaluated. Detection is 87–94% within a sub-second budget. At this testbed's probe economics the measured tier costs span only a factor 2.4, so cost-normalization separates the routing policies weakly; Section 6.5 isolates the routing question in a controlled cost regime.

6.2 Validity across the operating range

Sweeping $\alpha \in \{0.01, 0.02, 0.05, 0.10, 0.20\}$ (100 replications per level, Kelly routing) yields empirical false-rejection 0.0000 at *every* level ($n \approx 290$–300 faithful candidates per level), with detection stable at 92–94%. Two remarks. First, the guarantee of Theorem 2 is an upper bound, not a calibration claim: ECLAIR promises *at most* α, and the observed margin is the structural price of a composite null — tier B cannot false-alarm by proof, and tiers A/C bet against upper-bounded null rates because the true rates are unknowable. Second, the flatness of the detection curve is Proposition 4 in action: rejection budgets grow only logarithmically as α tightens.

6.3 Tier ablation: what each probe class buys

Table 2 repeats the testbed with restricted probe pools.

Three findings. *(i) Entanglement consumes the conservative margin.* Every configuration containing an unentangled tier records zero false rejections; tier C alone records 0.0401 — still within $\alpha = 0.05$, but with the margin that conservative calibration provides everywhere else fully spent. This is the empirical face of the Remark after Proposition 5: cross-model probes are safe as a supplement, and their calibration

Table 2 Probe-tier ablation (100 replications/configuration, Kelly, $\alpha = 0.05$).

Tiers	False rej.	Detection	Kill cost
A+B+C	0.0000	0.926	21 ms
A+B	0.0000	0.943	21 ms
A+C	0.0000	0.821	17 ms
B+C	0.0000	0.709	36 ms
A	0.0000	0.954	21 ms
C	0.0401	0.631	71 ms

must be conditioned on pool composition before they can carry certification alone. *(ii) Hard certificates matter beyond their rate.* Removing tier B (A+C vs. A+B+C) costs about ten points of detection although tier-B alarms occur on only ∼4% of mutant probes. *(iii) A caveat on tier A's dominance.* Tier A alone detects 95% here, but mutation screening guarantees every mutant is micro-instance-detectable — precisely tier A's regime. Errors of LLM origin need not be (conventions that diverge only at scale; see below), which is why the tiered design is retained.

6.4 End-to-end study with real LLMs

Thirty candidates were generated from natural-language specifications of the five families: two model families accessed through OpenRouter (a free-tier routed model and `nvidia/nemotron-3-ultra-550b`) × three prompt styles (direct; restate-then-model; expert persona) × per-style temperatures. Generated code is executed in a sandboxed namespace; an intake gate requires each candidate to build and solve two micro-instances. For scoring only, a proxy faithfulness label compares each candidate against the enumeration oracle on twelve micro-instances. All prompts, raw responses, and the response cache are included in the supplementary material, and this paragraph constitutes the documentation of LLM use in this study's methods.

Generation. 20/30 candidates passed intake. All ten failures were *loud*: API hallucinations importing conventions from other modelling ecosystems — `cp.Binary` and `cp.binary_var` (PuLP/DOcplex idioms), `cp.Var`, `cp.multiply`, `IntVar(domain=...)` (Choco-style), `Model().NewBoolVar` (raw CP-SAT API), a bare `import cp` — each caught by the gate at negligible cost.

Certification. All 20 intake survivors were proxy-faithful, so the certified pools are all-null: the correct behaviour is to reject nothing, and that is what happened — 0/20 false rejections under every routing policy, the certified pick was faithful in all five families, no abstentions, and no e-BH false discoveries. The guarantee of Theorem 2 thus survives contact with real generated models, complementing the mutation testbed in which the same machinery demonstrably detects 87–94% of actual faults.

Two honest observations shape the outlook. On well-pinned specifications, these two model families fail loudly (crashes) rather than silently (wrong models) — so the certification layer's detection power is insurance whose premium, by Proposition 4, is logarithmically cheap. And provoking the silent-error regime in LLM candidates requires deliberately ambiguous, convention-laden specifications (the colouring objective convention — "number of colours" vs. "maximum colour index" — is the template), which is exactly where ECLAIR's abstention-with-evidence output becomes the interesting object of study; we return to this in Section 7.

6.5 Routing under heterogeneous probe costs

The CP-SAT testbed's probes all cost single-digit milliseconds, which compresses the routing question. To isolate it, a controlled simulation reproduces the certification layer with the measured alarm-rate structure but a 20:1 probe-cost spread (expensive near-exact oracles vs. cheap noisy checks — the regime of industrial instances, of costly solver calls, and of LLM-as-judge probes priced per token). Over 500 replications: Kelly routing detects 86% of unfaithful candidates at mean cost 12.2 budget units per kill, against 60% at 16.6 for round-robin and 77% at 22.6 for cost-blind selection — while false rejection stays at 0.000–0.003, below α for every policy. Routing gains scale with probe-cost heterogeneity; validity is routing-invariant, as Theorem 2 requires.

7 Limitations and outlook

Calibration scope. Validity rests on Assumption 1; our evidence that conservative calibration transfers across problem families is empirical (zero false rejections under family transfer), not a theorem, and tier-C calibration should be conditioned on pool composition before cross-model evidence carries substantial weight — the ablation shows the margin, though never the bound, giving way. *Screened mutants.* The testbed's faults are guaranteed micro-detectable by construction; LLM-native silent errors need not be, and eliciting them requires deliberately ambiguous specifications — the planned abstention study, which also connects certification to conversational elicitation. *Checker provenance.* Our enumeration checkers are hand-written and independent; a fully LLM-generated pipeline must calibrate checker error as tier-A noise, which the framework accommodates but this study does not exercise. *Breadth.* Five families suffice to exercise every mechanism; scaling the probe library toward CSPLib/NL4Opt breadth, and stating the MR validity results for global constraints (`alldifferent`, `cumulative`) in the generality they deserve, are the natural next steps.

8 Conclusion

ECLAIR turns the verification of LLM-generated constraint models from a heuristic detection exercise into sequential hypothesis testing with explicit guarantees: anytime-valid false-rejection control under optional stopping and adaptive, budget-aware probe routing; FDR control across a generate–test–repair tournament; and an auditable evidence ledger for every decision, including abstention. On real CP-SAT probes the guarantee held exactly at every operating level while nine of ten injected faults were caught within sub-second budgets, and the framework certified real LLM-generated models without error. The broader claim is methodological: solver-in-the-loop verification of generative models is, at bottom, a sequential inference problem, and the e-process toolkit fits it exactly.

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Data availability. All code, calibration data, experimental results, and cached LLM generations are available in the supplementary material.

Use of large language models. LLMs are the object of study; their use in the experiments (models, prompting, decoding parameters, caching) is documented in Section 6.4.

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